Residual Representation and Quasi-interpolation

Part 1: residuals, patches, and the averaging operator

Lecture outline

We continue from elementwise integration by parts and prepare the residual identity for an error estimate.

1. Combine the element-boundary terms into interior and Neumann facet residuals.
2. Compute the element and nodal residuals on a nonuniform 1D mesh.
3. Use Galerkin orthogonality to subtract a finite element function.
4. Construct the Clément averaging operator and evaluate one example.

The local approximation estimates and the computable estimator are developed in the separate Part 2 reading deck.

Review: diffusion problem and test spaces

Recall the diffusion problem

$$\begin{aligned} -\nabla\cdot(k\nabla u) &= f \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \Gamma_D, \\ k\nabla u\cdot \boldsymbol{n} &= q \quad \text{on } \Gamma_N, \end{aligned}$$

Here q is the **prescribed normal flux** on Γ_N, and $\boldsymbol{n}$ is the outward unit normal:

$$\boldsymbol{q} = k\nabla u, \qquad q = \boldsymbol{q}\cdot\boldsymbol{n} = k\nabla u\cdot\boldsymbol{n}.$$

The test spaces are

$$\mathcal{V}_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}, \qquad \mathcal{V}_{h,0} \subset \mathcal{V}_0.$$

Here $\mathcal{V}_{h,0}$ is the continuous Lagrange space with zero values on Γ_D. Assume $|\Gamma_D| > 0$ and $0 < k_0 \leq k \leq k_1$ throughout.

Review: variational forms and energy norm

For $f \in L^2(\Omega)$ and $q \in L^2(\Gamma_N)$, define

$$a(w, v) = \int_\Omega k \nabla w \cdot \nabla v \, \mathrm{d}x, \qquad \ell(v) = \int_\Omega f v \, \mathrm{d}x + \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The continuous and discrete equations are

$$a(u, v) = \ell(v) \quad \forall v \in \mathcal{V}_0, \qquad a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in \mathcal{V}_{h,0}.$$

Since a is symmetric and coercive on $\mathcal{V}_0$, its energy norm is

$$\|v\|_a = \sqrt{a(v, v)} = \left(\int_\Omega k |\nabla v|^2 \, \mathrm{d}x \right)^{1/2}.$$

The same a and ℓ define both problems. The test space changes from $\mathcal{V}_0$ to its finite element subspace $\mathcal{V}_{h,0}$.

Review: residual and energy error

Since $u, u_h \in \mathcal{V}_0$, the error $e = u - u_h$ also belongs to $\mathcal{V}_0$. For any $v \in \mathcal{V}_0$, the continuous equation gives $\ell(v) = a(u, v)$. Starting from the definition of the residual,

$$\begin{aligned} R(u_h; v) &= \ell(v) - a(u_h, v) \\ &= a(u, v) - a(u_h, v) \qquad \text{(continuous equation)} \\ &= a(u - u_h, v) = a(e, v) \quad \text{(bilinearity)}. \end{aligned}$$

We may take $v = e$ because $e \in \mathcal{V}_0$. This gives

$$\boxed{R(u_h; e) = a(e, e) = \|e\|_a^2 .}$$

The last equality uses the definition $\|e\|_a = \sqrt{a(e, e)}$. Thus testing the residual with the error gives the squared energy error.

Review: integration by parts on each element

Start with the product rule on $\mathcal{K}_e$:

$$\nabla\cdot(vk\nabla u_h) = k\nabla u_h\cdot\nabla v + v\nabla\cdot(k\nabla u_h).$$

Integrate over $\mathcal{K}_e$ and apply the divergence theorem:

$$\int_{\partial\mathcal{K}_e} v(k\nabla u_h\cdot\boldsymbol{n}_e)\,\mathrm{d}s = \int_{\mathcal{K}_e} k\nabla u_h\cdot\nabla v\,\mathrm{d}x + \int_{\mathcal{K}_e} v\nabla\cdot(k\nabla u_h)\,\mathrm{d}x.$$

Rearrange to isolate the first volume integral:

$$\begin{aligned}\int_{\mathcal{K}_e} k\nabla u_h\cdot\nabla v\,\mathrm{d}x = &-\int_{\mathcal{K}_e}\nabla\cdot(k\nabla u_h)v\,\mathrm{d}x\\ &+\int_{\partial\mathcal{K}_e}(k\nabla u_h\cdot\boldsymbol{n}_e)v\,\mathrm{d}s.\end{aligned}$$

Here $\boldsymbol{n}_e$ is the outward unit normal of $\mathcal{K}_e$.

Expanding the residual before integration by parts

Start from $R(u_h; v) = \ell(v) - a(u_h, v)$. Substitute the definitions of ℓ and a:

$$R(u_h; v) = \int_\Omega fv \, \mathrm{d}x + \int_{\Gamma_N} qv \, \mathrm{d}s - \int_\Omega k \nabla u_h \cdot \nabla v \, \mathrm{d}x.$$

The elements partition Ω, so each volume integral is a sum of element integrals:

$$\begin{aligned} R(u_h; v) = &\sum_e \int_{\mathcal{K}_e} fv \, \mathrm{d}x + \int_{\Gamma_N} qv \, \mathrm{d}s \\ &- \sum_e \int_{\mathcal{K}_e} k \nabla u_h \cdot \nabla v \, \mathrm{d}x. \end{aligned}$$

Apply the preceding Green's formula to each integral in the second line. The minus sign in front of this sum changes both signs in that formula.

Element and boundary contributions to the residual

In the last line of the preceding slide, substitute Green's formula:

$$R(u_h; v) = \sum_e \int_{\mathcal{K}_e} \big(f + \nabla\cdot(k\nabla u_h)\big) v \, \mathrm{d}x \\ - \sum_e \int_{\partial\mathcal{K}_e} (k\nabla u_h \cdot \boldsymbol{n}_e) v \, \mathrm{d}s + \int_{\Gamma_N} q v \, \mathrm{d}s.$$

The first line is already separated by element. In the second line:

- an interior facet appears twice, once from each neighboring element;
- a Neumann facet appears once and combines with the prescribed-flux integral; and
- a Dirichlet facet contributes zero because $v = 0$ on Γ_D.

Interior facets: combining the two boundary terms

Let $\mathcal{F} = \partial\mathcal{K}^+ \cap \partial\mathcal{K}^-$. The outward normals are $\boldsymbol{n}_\mathcal{F}$ for $\mathcal{K}^+$ and $-\boldsymbol{n}_\mathcal{F}$ for $\mathcal{K}^-$. The two boundary terms are

$$
\begin{aligned}
&- \int_\mathcal{F} k^+ \nabla u_h^+ \cdot \boldsymbol{n}_\mathcal{F}\, v \, \mathrm{d}s - \int_\mathcal{F} k^- \nabla u_h^- \cdot (-\boldsymbol{n}_\mathcal{F})\, v \, \mathrm{d}s \\
&\qquad = - \int_\mathcal{F} \big(k^+ \nabla u_h^+ - k^- \nabla u_h^-\big) \cdot \boldsymbol{n}_\mathcal{F}\, v \, \mathrm{d}s.
\end{aligned}
$$

The superscripts identify values from the two elements. The same v appears in both terms because its values match across the shared facet. Define

$$
j_\mathcal{F} = -\big(k^+ \nabla u_h^+ - k^- \nabla u_h^-\big) \cdot \boldsymbol{n}_\mathcal{F}.
$$

If the element labels are exchanged, the new normal is $-\boldsymbol{n}_\mathcal{F}$. The new residual is

$$
-(k^- \nabla u_h^- - k^+ \nabla u_h^+) \cdot (-\boldsymbol{n}_\mathcal{F}) = j_\mathcal{F}.
$$

The flux difference and the normal both change sign, so their dot product is unchanged.

Boundary facets: Neumann mismatch and Dirichlet values

On a Neumann facet, combine the two boundary terms from the residual:

$$-\int_{\mathcal{F}}(k\nabla u_h \cdot \boldsymbol{n})v \, \mathrm{d}s + \int_{\mathcal{F}} qv \, \mathrm{d}s = \int_{\mathcal{F}}(q - k\nabla u_h \cdot \boldsymbol{n})v \, \mathrm{d}s.$$

The Neumann-facet residual is therefore

$$j_{\mathcal{F}} = \underbrace{q}_{\text{prescribed}} - \underbrace{k\nabla u_h \cdot \boldsymbol{n}}_{\text{computed}}.$$

On a Dirichlet facet, the test function satisfies $v = 0$, so

$$-\int_{\mathcal{F}}(k\nabla u_h \cdot \boldsymbol{n})v \, \mathrm{d}s = 0.$$

There is no Dirichlet-facet term in the residual representation.

Complete element-and-facet residual representation

Let $\mathcal{F}_h^{\mathrm{int}}$ denote interior facets and $\mathcal{F}_h^N$ boundary facets on Γ_N. The residuals are

$$r_e = f + \nabla\cdot(k\nabla u_h) \quad \text{on } \mathcal{K}_e.$$

$$j_{\mathcal{F}} = \begin{cases} -(k^+\nabla u_h^+ - k^-\nabla u_h^-)\cdot \boldsymbol{n}_{\mathcal{F}}, & \mathcal{F}\in\mathcal{F}_h^{\mathrm{int}}, \\ q - k\nabla u_h\cdot\boldsymbol{n}, & \mathcal{F}\in\mathcal{F}_h^N. \end{cases}$$

The residual is therefore

$$\boxed{R(u_h; v) = \sum_e \int_{\mathcal{K}_e} r_e v \,\mathrm{d}x + \sum_{\mathcal{F}\in\mathcal{F}_h^{\mathrm{int}}\cup\mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} v \,\mathrm{d}s.}$$

Each interior facet appears once in this exact identity.

What the two residuals measure

Element residual
$r_e = f + \nabla \cdot (k \nabla u_h)$
balance defect inside $\mathcal{K}_e$

Facet residual
$j_{\mathcal{F}}$
flux jump or mismatch

For P_1 elements and elementwise-constant k, ∇u_h and $k \nabla u_h$ are constant on each element. Hence

$$\nabla \cdot (k \nabla u_h) = 0, \qquad r_e = f|_{\mathcal{K}_e}.$$

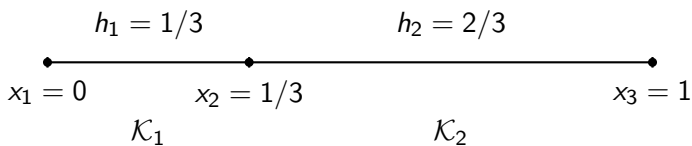
The element residual then reduces to the prescribed source. The computed-gradient information is contained in the facet residuals.

Nonuniform 1D example: mesh and computed solution

Consider the problem

$$-u'' = 1 \quad \text{on } (0,1), \qquad u(0) = 0, \qquad u'(1) = 0.$$

Here $k = 1$, $f = 1$, and $q = 0$ at $x = 1$. Take the computed nodal values $U_1 = 0$, $U_2 = 5/18$, and $U_3 = 1/2$ as given.



$$u_h'|_{\mathcal{K}_1} = \frac{U_2 - U_1}{h_1} = \frac{5/18 - 0}{1/3} = \frac{5}{6}, \qquad u_h'|_{\mathcal{K}_2} = \frac{U_3 - U_2}{h_2} = \frac{1/2 - 5/18}{2/3} = \frac{1}{3}.$$

Since $u_h'' = 0$ inside each element, $r_1 = r_2 = f + u_h'' = 1$. The remaining residuals are at the nodes.

The 1D residual includes nodal facet terms

In one dimension, facet integrals become evaluations at nodes. At x_2, the outward normals are $n_1 = +1$ from $\mathcal{K}_1$ and $n_2 = -1$ from $\mathcal{K}_2$. Substitute the computed slopes into

$$j_2 = -\big(u_h'|_{\mathcal{K}_1} n_1 + u_h'|_{\mathcal{K}_2} n_2\big) = -\left(\frac{5}{6}(+1) + \frac{1}{3}(-1)\right) = -\frac{1}{2}.$$

At $x_3 = 1$, the prescribed normal datum is $q = 0$, so

$$j_3 = q - u_h'(1)\,(+1) = 0 - \frac{1}{3} = -\frac{1}{3}.$$

Substitute $r_1 = r_2 = 1$, $j_2 = -1/2$, and $j_3 = -1/3$ into the residual representation:

$$\begin{aligned} R(u_h; v) &= \sum_{e=1}^{2} \int_{\mathcal{K}_e} r_e v \, \mathrm{d}x + j_2 v(x_2) + j_3 v(x_3) \\ &= \int_{\mathcal{K}_1} v \, \mathrm{d}x + \int_{\mathcal{K}_2} v \, \mathrm{d}x - \frac{1}{2} v(x_2) - \frac{1}{3} v(x_3). \end{aligned}$$

Here $v \in H^1(0,1)$ and $v(0) = 0$, so there is no term at $x_1 = 0$.

Constructing a residual estimator

Remove the unknown error from the residual identity, identify the mesh-size weights, and obtain a computable upper bound.

The unknown error in the residual identity

Combining $R(u_h; e) = \|e\|_a^2$ with the residual representation gives

$$\|e\|_a^2 = R(u_h; e) = \sum_{e'} \int_{\mathcal{K}_{e'}} r_{e'} e \, \mathrm{d}x + \sum_{\mathcal{F} \in \mathcal{F}_h^{\mathrm{int}} \cup \mathcal{F}_h^N} \int_{\mathcal{F}} j_{\mathcal{F}} e \, \mathrm{d}s.$$

Here e' indexes elements, while e is the error function. All residuals are known, but e is not. By Cauchy–Schwarz,

$$\left| \int_{\mathcal{K}_{e'}} r_{e'} e \, \mathrm{d}x \right| \leq \|r_{e'}\|_{L^2(\mathcal{K}_{e'})} \|e\|_{L^2(\mathcal{K}_{e'})}$$

which still contains the unknown local error norm.
The goal is a bound $\|e\|_a \leq C_{\mathrm{rel}} \eta$, where η uses only u_h, the data, and the mesh. A finite element approximation of e will let us introduce the local mesh sizes into this bound.

Testing the residual with the error

Both u and u_h satisfy the same zero Dirichlet condition. Hence $e = u - u_h \in \mathcal{V}_0$ is an admissible test function.
Take $v = e$ in the continuous equation $a(u, v) = \ell(v)$, giving $\ell(e) = a(u, e)$. Starting from the residual definition,

$$
\begin{aligned}
R(u_h; e) &= \ell(e) - a(u_h, e) && \text{(definition of the residual)} \\
&= a(u, e) - a(u_h, e) && \text{(continuous equation, } v = e\text{)} \\
&= a(u - u_h, e) && \text{(bilinearity in the first argument)} \\
&= a(e, e) && (e = u - u_h) \\
&= \|e\|_a^2 && \text{(definition of the energy norm).}
\end{aligned}
$$

Subtracting a finite element function

The preceding calculation gives $R(u_h; e) = \|e\|_a^2$. The discrete equation is $a(u_h, v_h) = \ell(v_h)$ for every $v_h \in \mathcal{V}_{h,0}$. It gives the separate identity

$$R(u_h; v_h) = \ell(v_h) - a(u_h, v_h) = \ell(v_h) - \ell(v_h) = 0.$$

Now use linearity of the residual in its test function:

$$\begin{aligned} R(u_h; e - v_h) &= R(u_h; e) - \underbrace{R(u_h; v_h)}_{0} \\ &= R(u_h; e) = \|e\|_a^2. \end{aligned}$$

We can choose v_h to approximate e without changing the residual value.

An approximation that does not need nodal values

For a continuous function v, the nodal interpolant from Chapter 7 is

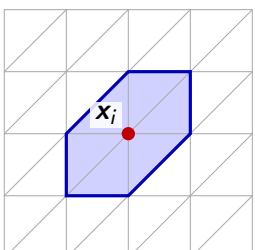
$$\mathcal{I}_h v = \sum_i v(\boldsymbol{x}_i)\phi_i.$$

But $e = u - u_h$ is known only to belong to $H^1(\Omega)$. In two and three dimensions, an H^1 function need not have well-defined point values $e(\boldsymbol{x}_i)$.
A local average is well defined even when a point value is not. We will replace $v(\boldsymbol{x}_i)$ by an average over the elements meeting at $\boldsymbol{x}_i$.
This defines a **quasi-interpolation operator** $\mathcal{Q}_h$. Its output is a continuous finite element function, so it can serve as v_h in the subtraction identity.

A vertex patch

A **vertex patch** ω_i contains all elements meeting at the vertex $\boldsymbol{x}_i$:

$$\omega_i = \bigcup_{\mathcal{K}_e:\, \boldsymbol{x}_i \in \overline{\mathcal{K}_e}} \overline{\mathcal{K}_e}.$$



The six blue triangles form ω_i in this mesh. The nodal basis function ϕ_i is nonzero only on these elements, so

$$\operatorname{supp} \phi_i = \omega_i.$$

The average used for coefficient i will use v on this blue region.

The Clément averaging construction

For P_1 elements, replace the nodal value by the **patch average**

$$c_i(v) = \frac{1}{|\omega_i|} \int_{\omega_i} v \, \mathrm{d}x, \qquad \boldsymbol{x}_i \notin \overline{\Gamma_D}.$$

Here $|\omega_i|$ is the area or volume of the patch. Set $c_i(v) = 0$ at Dirichlet vertices. With N_n mesh vertices, form

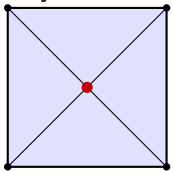
$$\boxed{\mathcal{Q}_h v = \sum_{i=1}^{N_n} c_i(v)\phi_i.}$$

The shared nodal coefficients make $\mathcal{Q}_h v$ continuous. On a Dirichlet facet, all vertex coefficients are zero, so the linear function is zero on the whole facet. Thus $\mathcal{Q}_h v \in \mathcal{V}_{h,0}$ for $v \in \mathcal{V}_0$.

Averaging version: Schöberl, §4.3, Definition 74, p. 64.

A concrete quasi-interpolant

Take $\Omega = (-1,1)^2$, $\Gamma_D = \partial\Omega$, and $v(x_1,x_2) = (1-x_1^2)(1-x_2^2)$. This function is zero on the boundary.



$\boldsymbol{x}_1 = (0,0)$
$\omega_1 = \overline{\Omega}$

The only free vertex is $\boldsymbol{x}_1$, with $|\omega_1| = 4$. First evaluate

$$\int_{-1}^{1}(1-t^2)\,\mathrm{d}t = \left[t - \frac{t^3}{3}\right]_{-1}^{1} = \frac{4}{3}.$$

The product form of v lets us factor the double integral. All boundary coefficients are zero.

$$\begin{aligned} c_1(v) &= \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1}(1-x_1^2)(1-x_2^2)\,\mathrm{d}x_1\,\mathrm{d}x_2 \\ &= \frac{1}{4}\left(\int_{-1}^{1}(1-t^2)\,\mathrm{d}t\right)^2 = \frac{1}{4}\left(\frac{4}{3}\right)^2 = \frac{4}{9}. \end{aligned}$$

Thus $\mathcal{Q}_h v = \frac{4}{9}\phi_1$, whereas $\mathcal{I}_h v = \phi_1$ since $v(0,0) = 1$.

Residual identity and continuation

For $e = u - u_h$, the residual satisfies

$$R(u_h; v) = a(e, v), \qquad R(u_h; v_h) = 0 \quad (v_h \in \mathcal{V}_{h,0}).$$

The averaging construction gives $\mathcal{Q}_h e \in \mathcal{V}_{h,0}$. Hence

$$\|e\|_a^2 = R(u_h; e) = R(u_h; e) - R(u_h; \mathcal{Q}_h e) = R(u_h; e - \mathcal{Q}_h e).$$

The error is still unknown. Local approximation estimates will bound $e - \mathcal{Q}_h e$ and introduce the mesh-size weights.

Part 2: local estimates, the residual estimator, and reliability.
Part 3: interpretation of the estimator and adaptive refinement.

References for the construction and estimates

- J. Schöberl, *Numerical Methods for Partial Differential Equations*, §4.3, pp. 64–67. Definition 74 gives the patch-average construction. Theorems 75–76 give its local estimates and reliability. Online version: §27.1–27.2.
- D. N. Arnold, *Lecture Notes on Numerical Analysis of Partial Differential Equations*, Chapter 4, §9.1–9.4, pp. 67–73. Equations (4.16)–(4.18) give local estimates for a projection-based Clément operator. Equations (4.19)–(4.21) derive the residual bound.
- M. G. Larson and F. Bengzon, *The Finite Element Method: Theory, Implementation, and Practice*, §4.9.1, pp. 88–89, Theorem 4.7. See also §2.6.1, pp. 38–39, for the 1D argument.

The references develop the homogeneous Dirichlet case. Our proof retains the Neumann residual and uses the boundary-preserving estimates stated in Chapter 8 as Hypothesis (Q).